DMart Retail Sales Analytics Dashboard (2025)

Author: Yash Sandeep Bhagyawant | **Type:** Interactive BI Dashboard (HTML, CSS, JavaScript) | **Dataset:** 2,500 cleaned transactions, 10 cities, 11 categories, 20 products. *Note: dataset is synthetic practice data, not actual DMart sales.*

1. Project Overview

Converted a raw retail dataset into a decision-ready dashboard covering Sales Performance, Profitability, Customer Behavior and Regional Trends, with cross-filtering for self-service Exploratory Data Analysis (EDA).

2. Data Cleaning & Preparation

- **Data Cleaning:** removed duplicates, handled missing/null values, corrected inconsistent entries.
- **Data Standardization:** consistent city, category, payment mode and order status labels.
- **Data Validation:** checked quantity, net sales, profit and discount for outliers and errors.
- **Feature Engineering:** derived Month, Net Sales, Estimated Profit and Discount fields.
- **Data Transformation & Modeling:** compact JSON structure; dimensions (city, category, channel, segment, payment, status, month) and measures (sales, profit, quantity, discount).

3. Key KPIs

Net Sales	Est. Profit	Profit Margin	Transactions	Units Sold	Avg Order Value
Rs 13.18 L	Rs 1.17 L	8.86%	2,500	7,876	Rs 527

4. Dashboard Components

- **Monthly Sales & Profit Trend** (combo chart): Time-Series and Seasonality Analysis.
- **Sales Channel, Payment Mode, Customer Segment, Order Status** donut charts: Omnichannel, Digital Payments, Customer Segmentation, Returns/Cancellation Analysis.
- **Category chart** (toggle Sales/Profit/Qty) and **City chart**: Category and Regional Performance Analysis.
- **Top 8 Products table:** Product Performance Ranking.
- **Interactivity:** 6 dropdown filters, Reset, click-to-cross-filter on every chart, sticky filter bar, responsive layout.

5. Key Insights

- Top city: Thane (~Rs 1.55 L), then Navi Mumbai and Nashik.
- Top categories: Home & Kitchen (~Rs 2.08 L) and Baby Care (~Rs 1.92 L); top product: Baby Diaper Pack.
- In-Store ~87% of sales vs Online ~13%: room to grow online.
- UPI leads payments (~36%). Regular customers ~57%, Members ~32%, Premium ~10%.
- Returns + cancellations ~Rs 76K (~5.8% of sales).
- Sales peak in Aug-Sep and Dec; dip in May.

6. Technical Skills Demonstrated

HTML5, CSS3, JavaScript (ES6), JSON, SVG, DOM Manipulation, Event Handling, Array Methods (filter, reduce, map, sort), Data Aggregation (Group By), CSS Grid & Flexbox, Responsive Design, KPI Design, Cross-Filtering, Drill-Down, Trend Analysis, Profit Margin Analysis, AOV, Dashboard Storytelling. No external libraries.

7. ATS-Friendly Resume Bullets

- Cleaned and standardized a **2,500-record retail transaction dataset** (deduplication, missing-value treatment, validation, feature engineering) to build a reliable analytics foundation.
- Designed and developed an **interactive sales dashboard** using **HTML, CSS and JavaScript** with 6 KPIs, 8 visualizations and 6 dynamic filters with cross-filtering.
- Performed **Exploratory Data Analysis (EDA)** across 10 cities, 11 categories and 5 payment modes, uncovering an **8.86% profit margin** and an **87:13 In-Store to Online sales split**.
- Delivered **actionable business insights** on regional performance, seasonality, product ranking and return/cancellation rates to support data-driven decision-making.

- Built **Time-Series, Category, Regional and Customer Segmentation analyses** with drill-down capability.

8. ATS Keywords

Primary: Data Analysis, Data Cleaning, Data Visualization, Dashboard Development, Business Intelligence, KPI Reporting, Exploratory Data Analysis (EDA), Retail Analytics, Sales Analysis.
Secondary: Data Preprocessing, Data Transformation, Data Validation, Feature Engineering, Trend Analysis, Seasonality Analysis, Customer Segmentation, Profit Margin Analysis, Regional Analysis, Product Performance Analysis, Cross-Filtering, Interactive Dashboards, Data Storytelling, Decision Support.
Technical: HTML, CSS, JavaScript, JSON, SVG, DOM, Responsive Design.